

Euler's relation: $e^{jx} = \cos(x) + j \sin x$

Convolution sum: $y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n]$

Convolution integration: $y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = x(t) * h(t)$

CT system function: $H(s) = \int_{-\infty}^{\infty} h(t)e^{-st} dt$

CT frequency response: $H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$

DT system function: $H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$

DT system function: $H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$

CT Fourier Series pair: $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t}$, $a_k = \frac{1}{T} \int_T x(t) e^{-jk\frac{2\pi}{T}t} dt$

CTFS conjugate symmetry: $x(t)$ is real $\Rightarrow a_{-k} = a_k^*$

CTFS time shift property: $x(t-t_0) \leftrightarrow a_k e^{-jk\frac{2\pi}{T}t_0}$

CTFS Parseval relation: $\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$

CTFS differentiation property: $\frac{dx}{dt} \leftrightarrow jk\omega_0 a_k$

CTFS integration property: $\int_{-\infty}^t x(t)dt \leftrightarrow (\frac{1}{jk\omega_0})a_k$

DTFS pair: $x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{N}n}$, $a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\frac{2\pi}{N}n}$

CTFT pair: $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$, $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

CTFT convolution property: $y(t) = x(t) * h(t) \leftrightarrow Y(j\omega) = X(j\omega)H(j\omega)$

	Time Function	Fourier Transform
1	$x(t) = \begin{cases} 1, & t \leq T_1, \\ 0, & t > T_1. \end{cases}$	$\frac{2 \sin(\omega T_1)}{\omega}$
2	$\frac{\sin(Wt)}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$
3	$\exp(-at)u(t), \quad a > 0$	$\frac{1}{a + j\omega}$
4	$t \exp(-at)u(t), \quad a > 0$	$\frac{1}{(a + j\omega)^2}$
5	$\delta(t)$	1
6	$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
7	$x(t) = 1$	$2\pi\delta(\omega)$
8	$\delta(t - t_0)$	$\exp(-j\omega t_0)$
9	$\exp(j\omega_0 t)$	$2\pi\delta(\omega - \omega_0)$
10	$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
11	$\sin(\omega_0 t)$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
12	$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - \frac{2\pi k}{T})$
13	$x(t) = \begin{cases} 1, & t \leq T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t+T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin(k\omega_0 T_1)}{k} \delta(\omega - k\omega_0)$
14	$\frac{t^{n-1}}{(n-1)!} \exp(-at)u(t), \quad a > 0$	$\frac{1}{(a + j\omega)^n}$
15	$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$